Structuration

Cours 5

Risque de Credit

ECP

Décembre 2025

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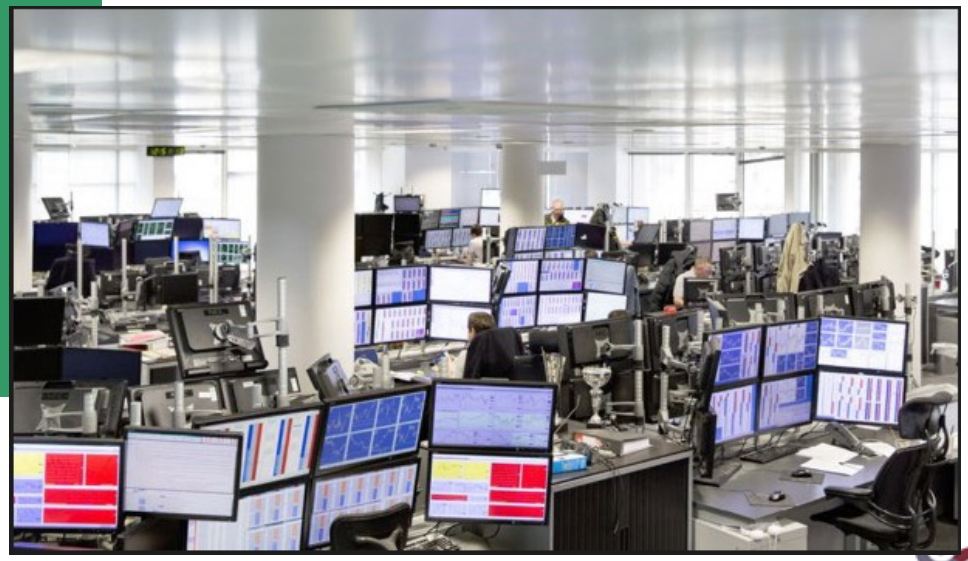

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Plan du cours

1. **Notions sur le MARCHÉ DU CREDIT**
   - Le Risque de crédit
   - Credit Linked Note
   - Repackaged Note

2. **Le risque de crédit dans les Dérivés**

3. **Exemple de distorsions de marché en temps de crise**
   - La crise des dettes souveraines
   - La gestion des ressources rares

LE RISQUE DE CREDIT

- **Credit Risk: Risk that an entity may experience a Credit Event over a time period**

- **Credit Events:**

  Corporate Names:

  - **Failure to pay**: The Reference Entity has failed to pay before the expiration of the grace period extension
  - **Bankruptcy**: The Reference Entity is insolvent, under liquidation or have decided to make an arrangement for the benefit of its creditors
  - **Restructuring**: Change in the agreement between the Reference Entity and the holder of an obligation due to a deterioration in creditworthiness or financial condition of the Reference Entity. This changes can be: reduction of interest or principal, postponement of payment of interest or principal, change in the ranking of priority of payment of an obligation, change of currency... (

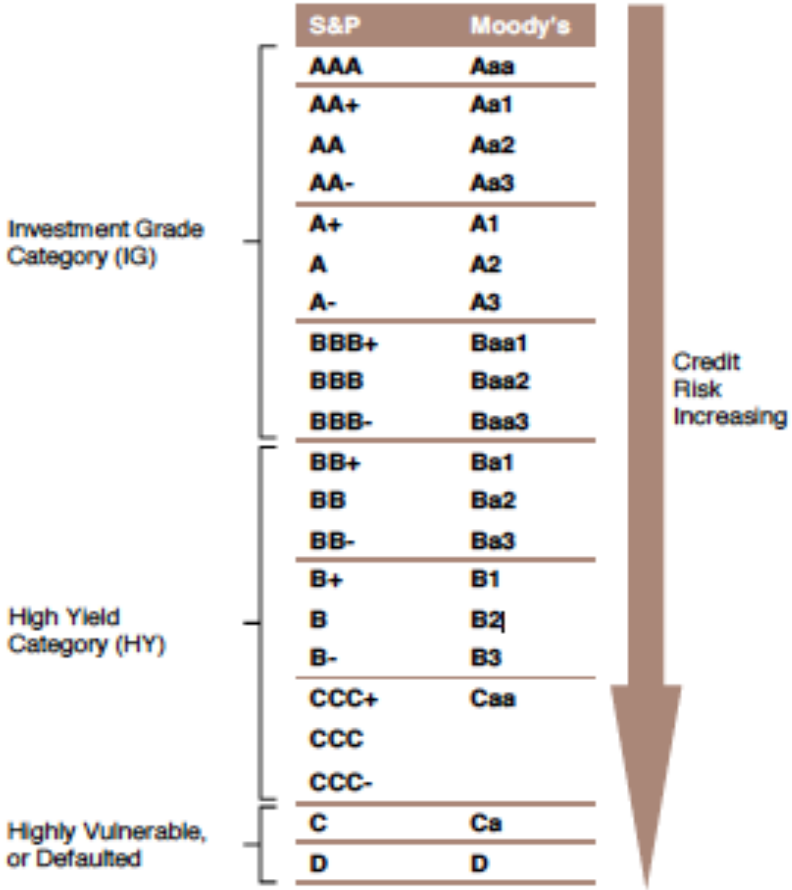
  Sovereign Names depending of the region, some of the following:

  - **Failure to pay**
  - **Restructuring**
  - **Repudiation/Moratorium**: An authorized officer or a governmental Authority rejects the validity of an Obligation, or impose a moratorium and failure to pay occurs

ANALYSE DU RISQUE DE CREDIT

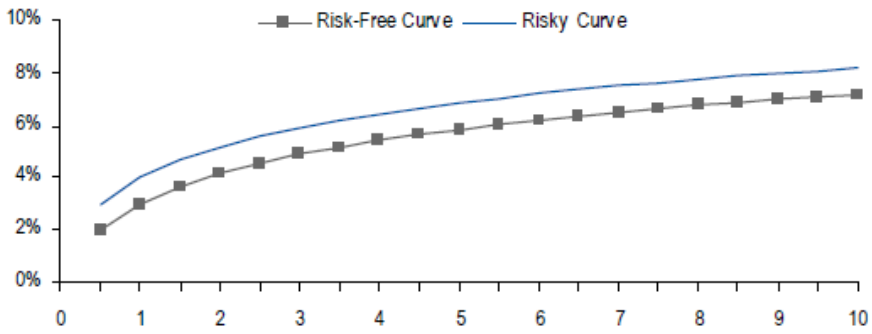
- **Le risque de crédit peu être analysée qualitativement par le biais de la notation (rating)**
  - 3 principales Rating Agencies : Standards & Poors, Moody's and Fitch
- **Les paramètres majeurs du risque de crédit sont :**
  - La probabilité de defaut
  - La perte, étant donné le defaut (LGD ou Loss Given Default equivalente à 100% - Recovery)

- **Ce risque de crédit est quantifiable en et la différence de rémunération de l'obligation et le niveau des taux sans risque (…) peut s'analyser au travers de 4 composants :**
  - **Risk-Free rate**: The bond holder could earn this yield in a default/risk-free investment (OAT, US Treasury rate).
  - **Swap Spread**: correspond à l'écart entre le yield de l'obligation Risk Free et le taux swap
  - **Credit Risk Premium**: The spread to compensate for the risk that the company defaults and investors lose future interest and principal payments.
  - **Bond Liquidity Premium**: The spread to compensate investors for the illiquidity of the bond.



MESURE DU RISQUE DE CREDIT

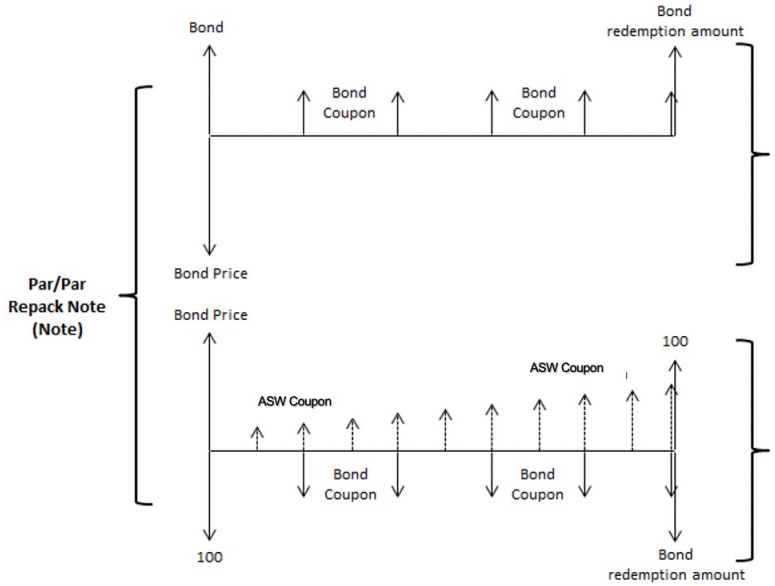
- **Credit and liquidity risks are measured by traditional "bond spread measures" such as Z-spread and asset swap spread.**



  - **Z-Spread:** The Z-spread is the parallel shift applied to the zero curve in order to equate the bond price to the present value of the cash flows.

$$\text{Dirtyprice} = \sum_{i=1}^{n} \frac{C}{[1+(r_i+z)]^i} + \frac{1}{[1+(r_i+z)]^i}$$

- **Asset Swap :**
  - An asset swap is similar in structure to a plain vanilla swap with the key difference being the underlying : a Bond
  - Rather than regular fixed and floating loan interest rates being swapped, fixed and floating assets are being exchanged.
  - The investor pays fixed and receives floating. This transforms the fixed coupon of the bond into a EURIBOR/LIBOR-based floating coupon.
  - It is widely used by banks to convert their long-term fixed rate assets to a floating rate.



  - **Asset Swap Spread**: The spread over Libor received on the floating side is called the asset swap spread, and can be considered to give some measure of the bond's credit risk.

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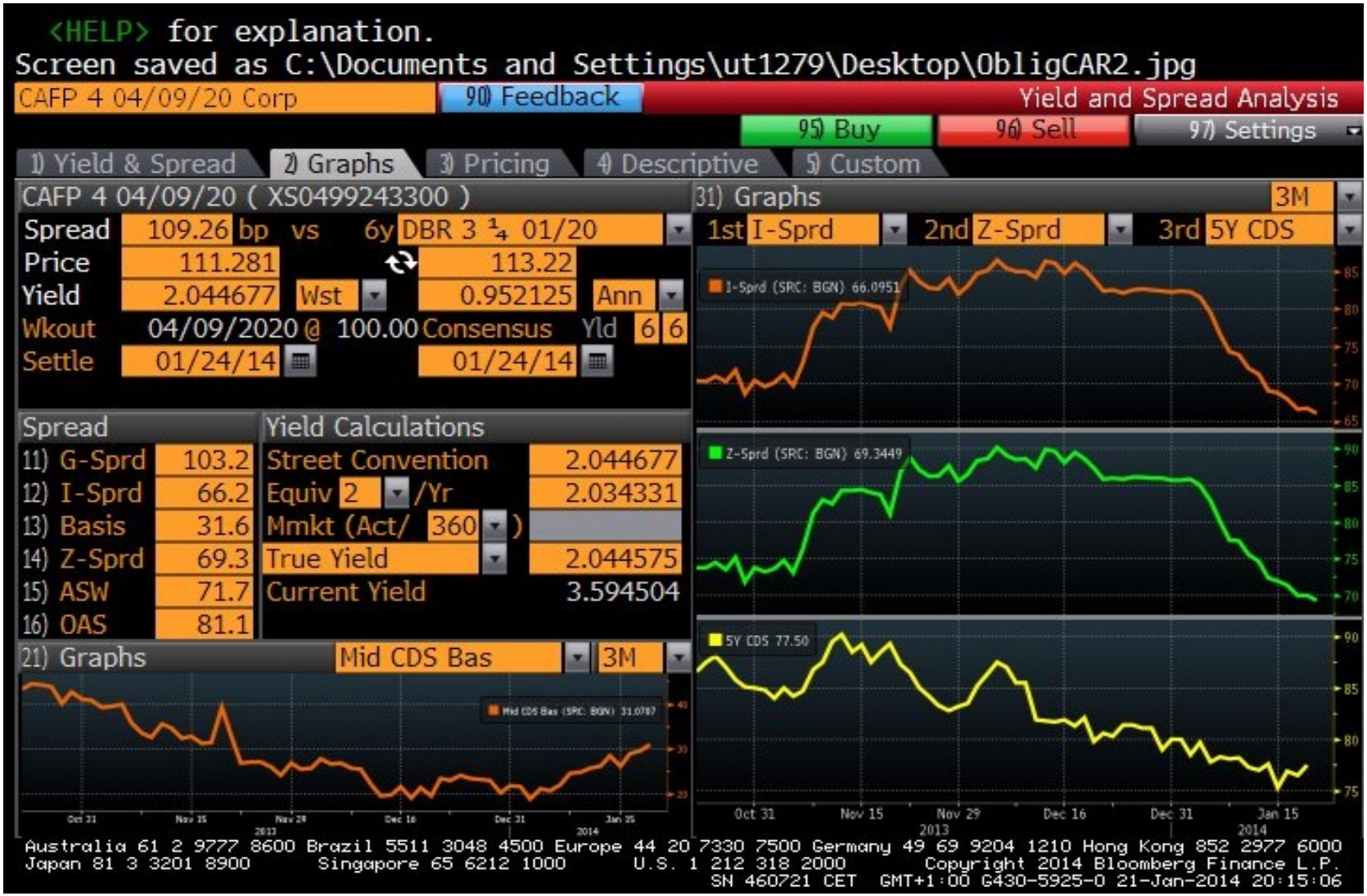
Exemple Analyse d'une Obligation



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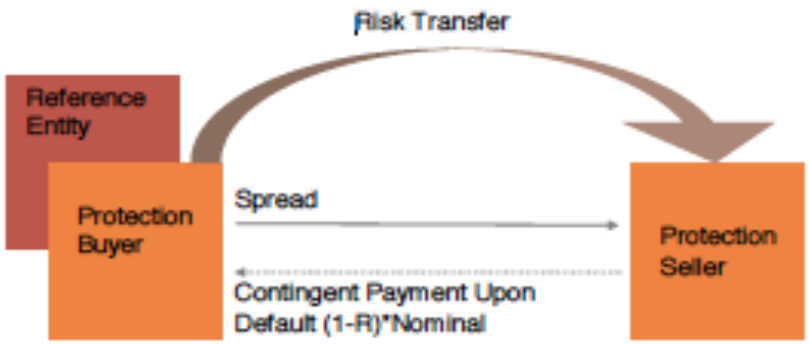
LES CDS

- **Objective:**
  - The objective of Credit Derivatives transactions (like CDS) is to isolate and separately transfer a Reference Entity's Credit Risk between two counterparties (i.e. bankruptcy and other events defined previously which materially impact on the credit quality of the Reference Entity's Obligations) => it is an insurance contract
- **Counterparties:**
  - Credit Derivatives are also known as "protection". Transactions in the market are usually referred to in terms of either buying or selling protection:
  - Buyer of Protection is seller of Credit Risk
  - Seller of Protection is buyer of Credit Risk
- **The CDS enables the Protection Buyer to hedge themselves against the Credit Risk of a Reference Entity by transferring this risk to the Protection Seller**
  - In return for this protection, the Protection Buyer pays a risk premium (the CDS SPREAD x Nominal) to theProtection Seller
  - The main characteristics of this contract are:
    - Trade Date
    - Effective Date
    - The Maturity
    - The Nominal
    - The debt subordination



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CDS Basis Definition

- **The Basis is the difference between the CDS and the ASW:**
  - Basis = CDS – ASW, also referred to as Basis-to-Cash.
  - In stable market conditions, the CDS and the spread should be similar as they both reflect market perception of the credit risk but there are:
    - Structural differences due to the nature of both instruments
    - Differences in the markets conditions

- **Liquidity is the most important factor**
  - Liquidity is driven by demand/supply dynamics
  - Especially in market stress CDS basis can move to extreme territories
  - In the coronavirus crisis, liquidity in the market has deteriorated : this is reflected mainly in physical bonds as opposed to CDS indices which are usually far more liquid.
  - In addition, the exceptionally large primary market activity has further penalised physical markets which had to absorb this exceptionally large supply.

Drivers	Effect on Basis
Bond issuance in illiquid and deteriorating credit conditions	Negative Basis
Bond issuer call options	Negative Basis
Bond repo / Funding costs	Negative Basis
Higher CDS relative liquidity (tightening spreads)	Negative Basis
Issuance of synthetic structured products	Negative Basis
Bond covenants protecting bond holders	Positive Basis
Cheapest-to-deliver option	Positive Basis
Higher CDS relative liquidity (widening spreads)	Positive Basis
Soft Credit Events	Positive Basis
Unwind of synthetic structured products	Positive Basis

- **Spread CDS > Spread ASW**
  (Basis >0)
- **Spread CDS < Spread ASW**
  (Basis <0)

CDS Basis Example

ENELIM 1.966 01/27/25 Corp | Settings | Yield and Spread Analysis

109.191/109.499 0.104/0.045 BGN@ 17:59 94) No Notes 95) Buy 96) Sell

1) Yield & Spread | 2) Graphs | 3) Pricing | 4) Description | 5) Custom

ENELIM 1.966 01/27/25 (XS1176079843)

Spread	70.45 bp vs	DBR 1 08/15/24	
Price	109.345	107.466	19:13:20
Yield	0.074254 Wst	-0.630287	Ann
Wkout	01/27/2025 @ 100.00	Consensus	Yld 6 6
Settle	02/14/20	02/14/20	

Spreads		Yield Calculations	
11) G-Sprd	67.9	Street Convention	0.074254
12) I-Sprd	35.0	Equiv 2 /Yr	0.074241
13) Basis	16.3	Mmkt (Act/360)	
14) Z-Sprd	35.3	True Yield	0.074253
15) ASW	37.1	Current Yield	1.798

Graphs 2Y

5Y CDS | ASW | ASW

Track Annotate Zoom

5Y CDS 52.12

150 100 50

ASW (SRC: BGN) 36.3526

100 50

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Basis Trade Opportunity

Basis trades exploit the different pricing of bonds and CDS on the same underlying company: by taking opposite positions in a bond and CDS, investors can profit from changes in the Bond-CDS basis

- **Rationale :**
  - Lock-In “Risk-free” Spread.
  - Trade the Basis.

- **Negative Basis Trade:**
  - The CDS spread is lower (tighter) than the bond spread
  - Buy the bond and buy CDS protection
  - Potential Profit from Default if the recovery is higher on the bond than on the CDS.

- **Positive Basis Trade :**
  - The CDS spread is higher (wider) than the bond spread
  - Borrow and short the bond (if possible) and sell CDS protection (long risk) with the same maturity (or as near as possible) as the bond
  - Investor needs to repo the bond and sell CDS protection. The difficulty with repo of corporate bonds and the cheapest-to deliver option that protection buyers own makes positive basis packages more difficult to analyse and execute.

STRIPPING

- **Le stripping de la courbe de crédit est obtenu par intégration, les défaut pouvant intervenir à tout instant**
- **Pour simplifier, l'exemple ci-dessous montre le principe du stripping en supposant que le « settlement » des CDS se fait annuellement et sans délais**

classiquement tires de la courbe des swaps

LGD		**40,0%**			
Maturité	**0d**	**1y**	**2y**	**3y**	**4y**
Niveau du CDS		2,00%	2,25%	2,50%	2,75%
Discount Factor		96,00%	92,16%	88,47%	84,93%
Proba de defaut sur l'année		4,76%	5,94%	7,17%	8,48%
proba cumulée de survie en fin d'année	100,0%	95,24%	89,58%	83,15%	76,10%

obtenu en égalisant la valeur actualisée des flux payés au titre du CDS :

2% x DF1y x (1- PD1y)

et la valeur des pertes attendues:
LGD x PD1y

Obtenus par itération, sachant PD1y et PS1y, en égalisant la valeur actualisée des flux payés au titre du CDS :

2,25% x DF1y x (PS1y) +
2,25% x DF2y x (PS1y) x (1-PD2y)

et la valeur des pertes attendues:
LGD x PD1y x DF1y +
LGD x PS1y x PD2y x DF2y|

LES INDICES DE CREDIT

- **The main credit indices are the Markit CDX and the Markit iTraxx:**
  - CDX Indices are linked to the U.S. credit market
  - iTraxx Indices are linked to the European credit market
- **The theoretical level of the Index is the duration weighted average of the CDS levels of each name. For example, here are some of the main iTraxx Indices**

- iTraxx Benchmark Indices: portfolios made of diversified European names from all industrial sectors

iTraxx Europe « Main »	iTraxx Europe HiVol	iTraxx Europe Crossover
125 most liquid European *Investment Grade* CDS names	Top 30 highest spread non-financial names from iTraxx Europe	45 European *High Yield* reference names

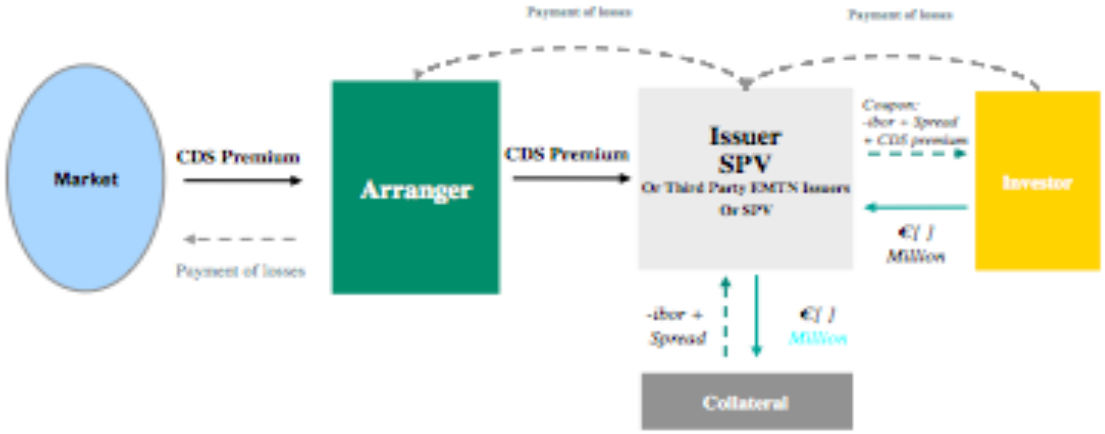
- iTraxx Sector Indices:
  - Portfolios of names chosen among the following sectors: Non-Financials (Autos & Industrials, Consumers, Energy, TMT sub-sectors composed of 100 entities), Financial Senior (25 entities), Financial Sub (25 entities).

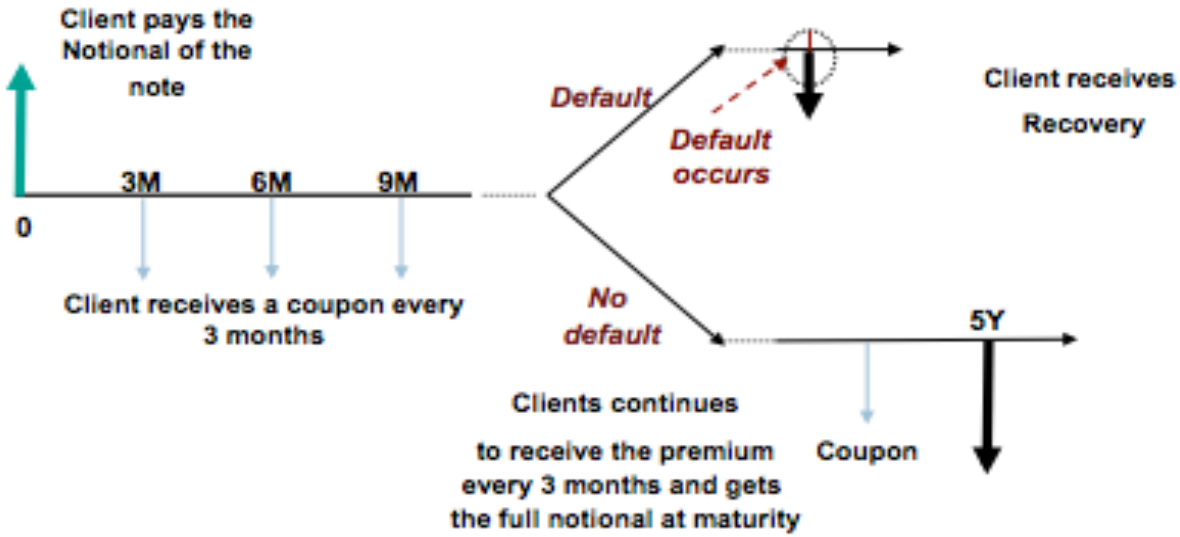
- The iTraxx indices trade 3, 5, 7 and 10-year maturities , 5y and 10y being the most liquid. The iTraxx Non-Financials, Financial Senior and Financial Sub indices trade only 5y and 10y.
- iTraxx Indices roll twice a year, on the 20th of March and on the 20th of September. Their underlying portfolio is updated. The current series are the series 11 for iTraxx index and series 12 for the CDX index
- For more information, see www.markit.com/

Credit Linked Note – Basic Mechanism

- **A Credit Linked Note (CLN) is a bond that synthetically replicates the performance of a Reference Entity.**
  - It is the funded version of a Credit Default Swap (A CLN is a Note backed by a deposit and a CDS on the underlying reference entity)
  - If the Reference Entity triggers a Credit Event, the Swap Counterparty (CA-CIB) receives Par minus Recovery, and the Investor the Recovery minus the hedge unwind costs (if applicable)

- **Applicability of ISDA 2003 Documentation & 2009 Supplements to ensure maximum liquidity and standardization**





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Credit Linked Note : Example

Emetteur	Crédit Agricole CIB Financial Solutions Garant Crédit Agricole CIB S&P LT : A ; Moodys : A1 ; Fitch : A+
Instrument	EMTN de droit Français
Maturité	**Selon maturité du sous-jacent**
Capital garanti	**Non** Capital remboursé à 100% si aucun évènement de crédit n'a lieu sur la durée du placement Sinon risque de perte en capital
Capital investi	**A partir de 500 000 EUR**
Caractéristiques du coupon et risque sous-jacent	

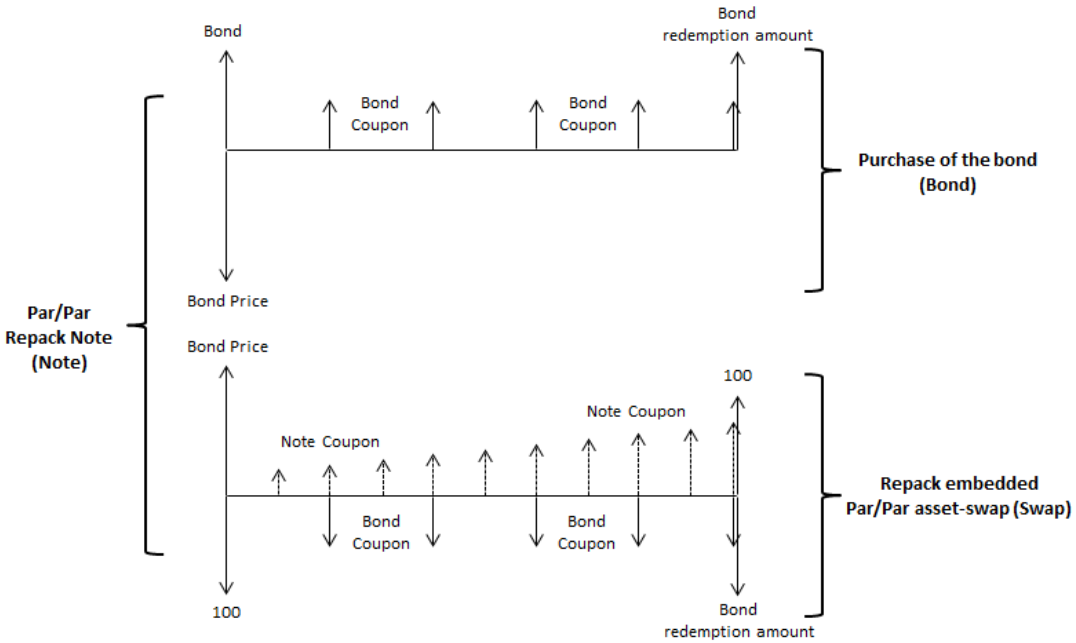
Entité sous-jacente	Ccy	Maturité	Paiement	Coupon payé	Rating
Veolia	EUR	30/06/2024	annuel, 30/360	0.65%	Moodys: Baa1; Fitch: BBB; S&P: BBB
Engie SA	EUR	30/06/2024	annuel, 30/360	0.66%	Moodys: A2 ; Fitch: A ; S&P : A-
AXA sub	EUR	30/06/2024	annuel, 30/360	1.15%	Fitch: A ; S&P : A
Veolia	EUR	30/06/2026	annuel, 30/360	0.92%	Moodys: Baa1; Fitch: BBB; S&P: BBB
Engie SA	EUR	30/06/2026	annuel, 30/360	0.92%	Moodys: A2 ; Fitch: A ; S&P : A-
AXA sub	EUR	30/06/2026	annuel, 30/360	1.61%	Fitch: A ; S&P : A
Portugal	EUR	30/06/2026	annuel, 30/360	1.25%	Moody's : Baa3; Fitch: BBB; S&P : BBB

Remboursement en capital, à l'échéance	100% à maturité si aucun évènement de crédit n'a eu lieu [Recouvrement – coût de couverture] sinon
Evènements de crédit	Définis par la Matrice ISDA (ISDA 2014 Credit Derivatives Définitions et ses suppléments) comme: (i) la faillite (ii) le défaut de paiement ou (iii) la restructuration de la dette (iiii) une répudiation ou un /moratoire

Repackaging – How it works

- **A repack note is a note issued by a SPV which:**
  - Has on its assets, the repack underlying bond (Bond)
  - Has on its liabilities, the Bond Repack Note (Note) issued by the SPV and purchased by the investor
  - Enters into a swap used to transform the Bond coupons and Bond redemption amount against the Note coupons and Note Redemption amount with the bank (swap counterparty) structuring the issuance of the repack note (Swap)

- **Repack flows in case of credit event:**



- **Repack flows in case of credit event:**
  - In case of a credit event, the note is early redeemed at:

Note Recovery Value =

Max(0, Bond Recovery Value + MTM (Swap))

  - The Swap counterparty (Bank) is senior on the underlying bond recovery under the swap Mark-To-Market (MTM)
  - The Note Recovery Value is floored at 0%
  - This floor is priced and usually called Gap Risk

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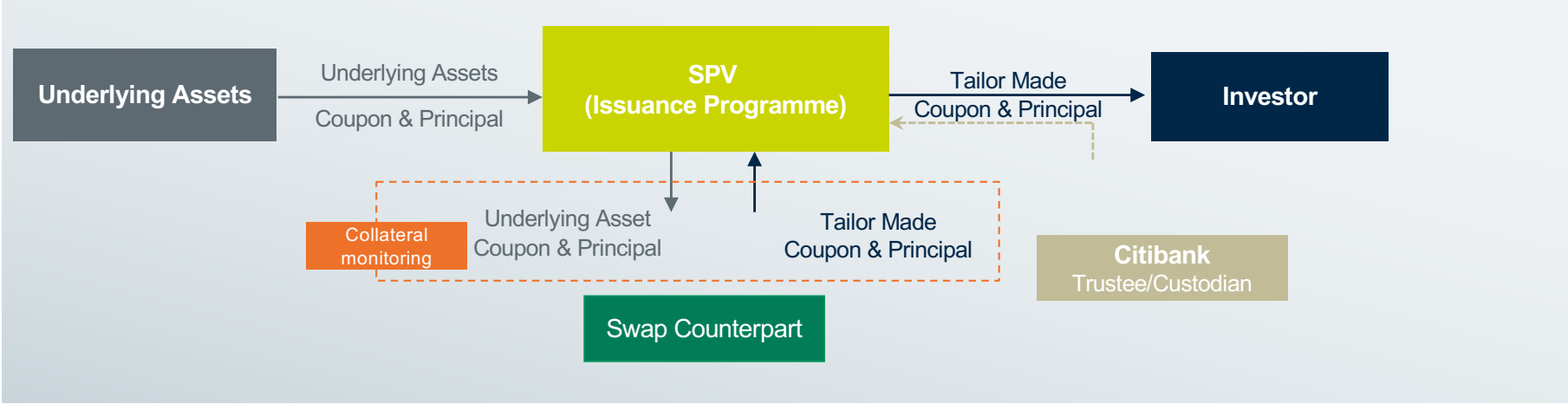
Repackaging – How it works



- **A repack note is a note issued by a SPV which:**
  - A SPV is a bankruptcy remote vehicle which issues a note
  - The proceeds of the note are invested in an asset of the investor's choice (ie. the collateral of the SPV). It can be a bond, a loan, etc.
  - The asset is put into an independent custodian that secures it for the benefit of the Noteholder. A trustee is also appointed (to defend all parties' best interest).
  - Eventually the SPV enters into a swap to transform the asset coupons / dividends into bespoke coupons / dividends as required by the investor (the "Asset Swap"). The asset swap can be collateralised through a CSA.
  - Economically the investor has the risk and reward of the asset and the asset swap.
  - In case of Credit Event in respect of the asset or swap counterparties, the asset and asset swap are unwound at market value and the proceeds (if any) are given back to the note holder.

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Example : OATei linker Repackaging: Fixed rate to CMS-linked note

Rationale of the transaction

- Tier 1 Insurance company looking to reduce its exposure to lapse risk in a rising long term rates environment

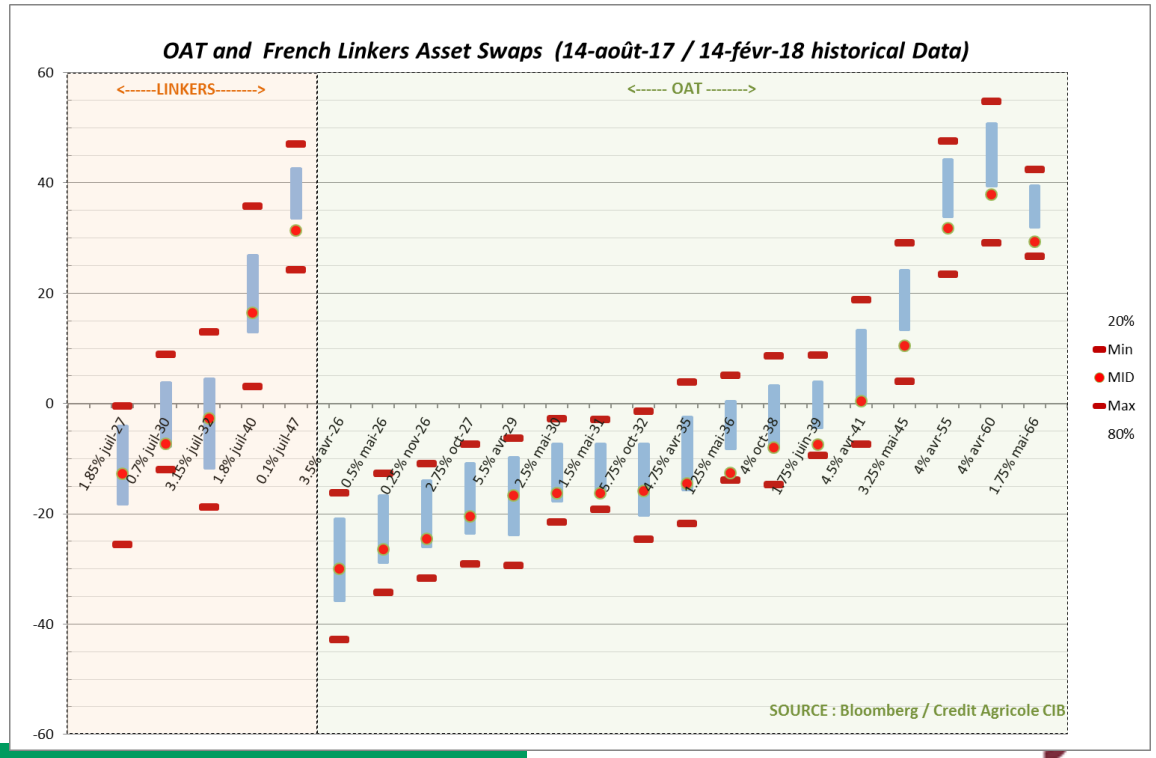
Specific requirements :

- Investment solution credit exposure: AA or above
- Investment solution providing 0% SCR credit
- Investment solution not exposing investor to a counterparty risk on the swap
- Issuance price at Par

Maturity	22.25Y
Deal Size	EUR 50 M
Format	Note issued by a Special Purpose Vehicle
Underlying Asset	The Note is secured by an Inflation-linked French Government Bond FRTR 1.80% July 2040
Issue price	100%
Redemption	100%
Coupon (PA30360)	Y1-2: 1.00% Y3+: EUR CMS20Y-65bp floor @0% and cap @8%
Reference#	Z-Spread 3bp

Solution

- Identification of a market opportunity in the bond market :
  - Long term French government bond trading wider than French agencies of similar credit
  - Inflation linker trading wider than nominal
- Swap Counterparty to fund the difference between OATei's dirty price and par



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Plan du cours

1. **Produits Structurés (Suite)**
   - Wrappers
2. **Notions sur le MARCHÉ DU CREDIT**
   - Le Risque de crédit
   - Credit Linked Note
   - Repackaged Note
3. **Le risque de crédit dans les Dérivés**
4. **Exemple de distorsions de marché en temps de crise**
   - La crise des dettes souveraines
   - La gestion des ressources rares

LE RISQUE DE CRÉDIT DANS LES DÉRIVÉS

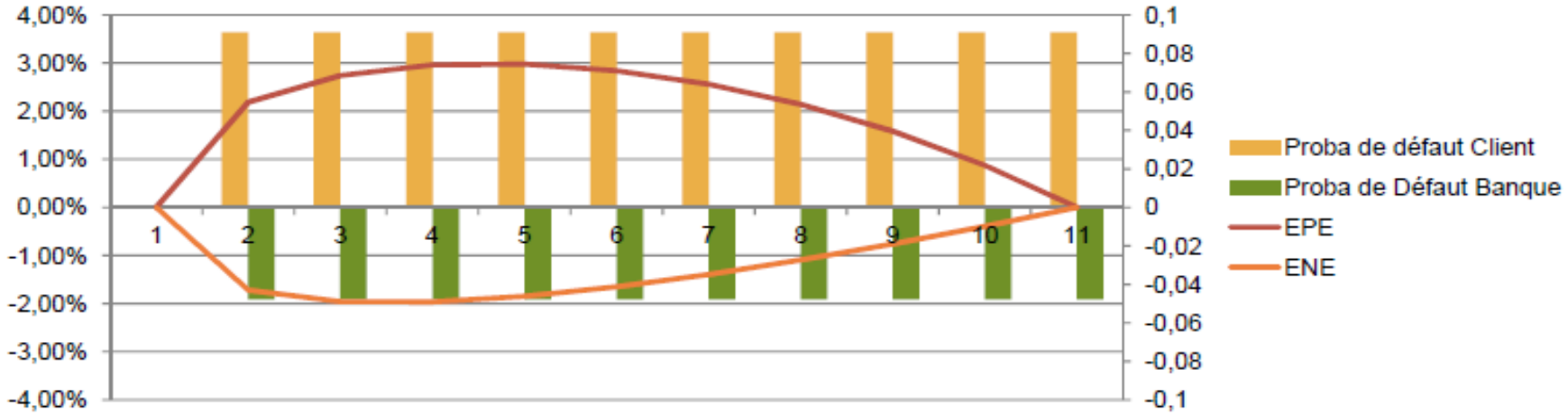
- **Un produit dérivé est un contrat ou deux parties s'engagent à échanger des flux à une date future.**

- **La valeur actualisée de ces flux est le Marked to Market de la transaction**
  - Si l'une des parties est en défaut au cours de la vie de la transaction et que le MTM est positif en sa faveur, l'autre contrepartie fera une perte égale à :
    - Le MTM de la transaction
    - Moins
    - La « recovery » sur ce MTM, c'est-à-dire le montant recouvré après liquidation des actifs de la société.
- **Si l'une des parties est en défaut au cours de la vie de la transaction mais que le MTM est négatif, ce MTM sera payé intégralement**

- **La banque qui rentre dans la transaction prend donc du risque de crédit sur le client / la contrepartie de l'opération**
- **Le nominal en jeu est variable et va dépendre du sous jacent**

LA CVA ET LA DVA

- **Le risque de crédit modifie la valeur de marché d'un dérivé.**
- **En pratique le vrai MTM doit prendre en compte la probabilité d'occurrence des flux**
  - En prenant en compte la probabilité de survie du client
  - En prenant en compte la probabilité de survie de la banque
- **La CVA (credit value adjustment) et la DVA (debit value adjustment) sont les noms courants des deux ajustements nécessaires**
- **La DVA est la CVA vue par le client.**
- **Le seul prix permettant que les deux contreparties puissent traiter la transaction est MTM hors crédit – CVA + DVA.**



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CVA – Definition

- The Credit Value Adjustment (CVA) is the **Counterparty Risk** embedded in every derivative trade

$$PV_{risky(credit)} = PV_{risk\ free} - CVA$$

- CVA (unilateral version) can be assumed (assuming spread / Recovery / exposure independence) as the amount of discounted future Expected Losses

$$CVA = \underbrace{LGD}_{\text{Loss Given Default}} * \sum_{t=0}^{Maturity} \underbrace{PD_{t-1,t}}_{\text{Default Probability}} * EPE_t * \underbrace{DF_t}_{\text{Exposure at Default (Discounted)}}$$

- For a typical (centralised) CVA Function
  - $PV_{risk\ free}$ – Hedged by Trading Desks
  - CVA – Hedged by CVA Trading Desk

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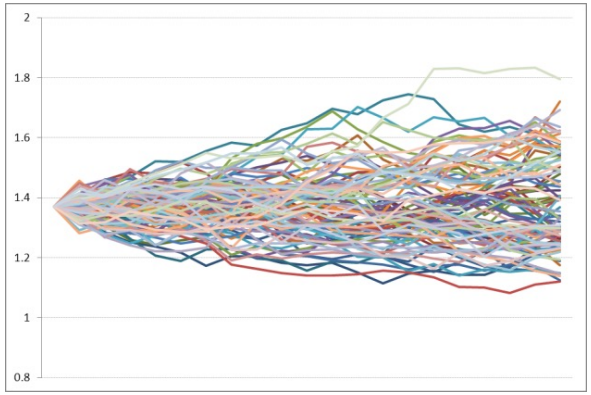
CVA – Key Ingredients

Expected Positive Exposure (EPE)

- Calculated via simulations process (Monte Carlo…)
- Computation including netting and collateral agreements
- Involves only the **Positive Exposures** in case of Counterparty Default
- Definition of Exposure linked to the mark to market of transaction
- Evaluated **Contingent** on the default of the counterparty
  - including right way / wrong way risks



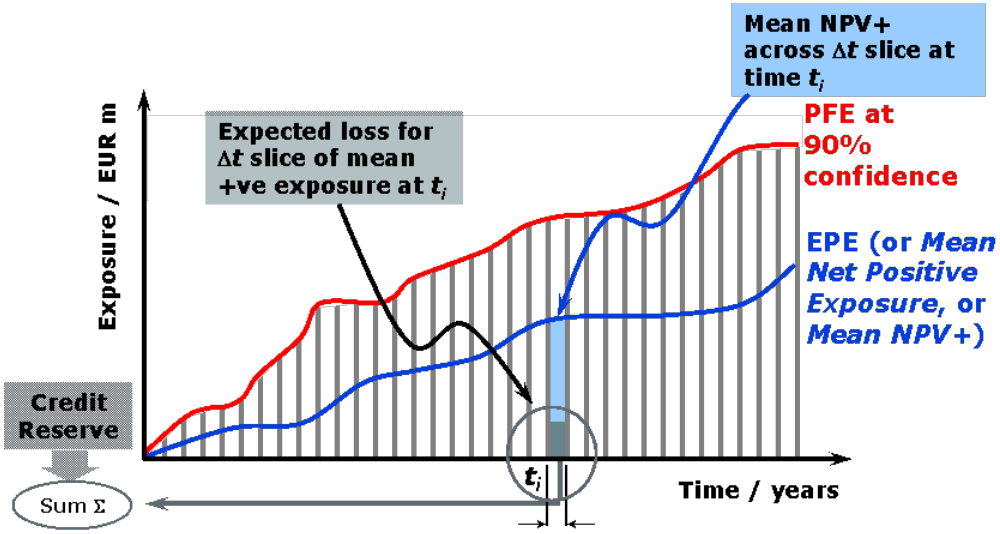
$$CVA = LGD * \sum_{t=0}^{Maturity} PD_{t-1,t} * EPE_t * DF_t$$

Default Probability

- Implied from CDS spreads (market-implied) or,
- Historical default probabilities

Loss Given Default / Recovery Rate

- Market Implied (where possible) : LGD_{Market}
- Internal Recovery measure : $LGD_{Internal}$



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GESTION DU RISQUE DE CREDIT

- **Le risque de crédit est géré par le biais d'ISDA, de CSA ou de Break Clause.**
- **Avant de traiter un dérivé, les contreparties signent en général un ISDA, contrat cadre qui permet :**
  - De définir les termes utilisés dans les contrats signés ensuite pour chaque deal
  - De gérer certains événement, notamment les règles de terminaison par anticipation en cas d'évènement spécifique (illégalité, changement fiscal, retenue à la source, défaut, faillure to pay)
  - Le **close out netting** entre les différents deals traitées sous cet ISDA, c'est-à-dire que les MTM positifs seront compensés des MTM négatifs au moment du défaut.
- **Le CSA (crédit support annexe) gouverne la gestion du collateral. Il impose le paiement d'appels de marge**
  - Par les deux contrepartie (bilatéral) ou par une seul contrepartie (Unilatéral)
  - En cash ou en titres
  - Avec une fréquence variable, souvent d'un jour, 1 semaine ou 1 mois
  - Dés le départ ou sur la base de « ratting triggers »
  - Un CSA en titre prévoit la rétrocession des coupons, un CSA en cash prévoir des intérêts, souvent à Eonia ou OIS

- **L'ISDA a un impact majeur :**
  - Le risque de crédit d'une opération ne peut pas se calculer sans prise en compte du risque de crédit des autres opérations
  - Il faut donc calculer la variation de CVAR/EPE du portefeuille total après inclusion de la nouvelle transaction.
  - Par exemple, sur un portefeuille comprenant un unique swap payeur de taux
    - Un swap receveur viendra réduire le CVAR et l'EPE
    - Un swap payeur viendra les augmenter
    - Un Xcurrency viendra augmenter mais apportera de la diversification.

- **Un CSA a deux impacts majeurs :**
  - Il réduit le risque de crédit, qui n'est plus qu'un risque de variation quotidienne / hebdomadaire / mensuelle du MTM
  - Il supprime les problématiques de liquidité
  - Les Break clause, qui impliquent une possibilité contractuelle de terminer le contrat par anticipation à la main des deux parties, permettent de réduire la maturité du risque de crédit

Credit Support Annex - CSA

Standard CSA terms

- **Bilateral margin transfer form:** symmetric transfer terms for the both parties
- **Daily margin call frequency** (weekly can be also accepted as standard)
- **No threshold** – up to 5M threshold is considered as reasonable
- **Cash and G7 bonds** (haircuts – 0%-2% on short term maturities and 5-10% on longer term maturities)
- **In EUR or in USD, remunerated at EONIA FLAT or Fed Funds FLAT**

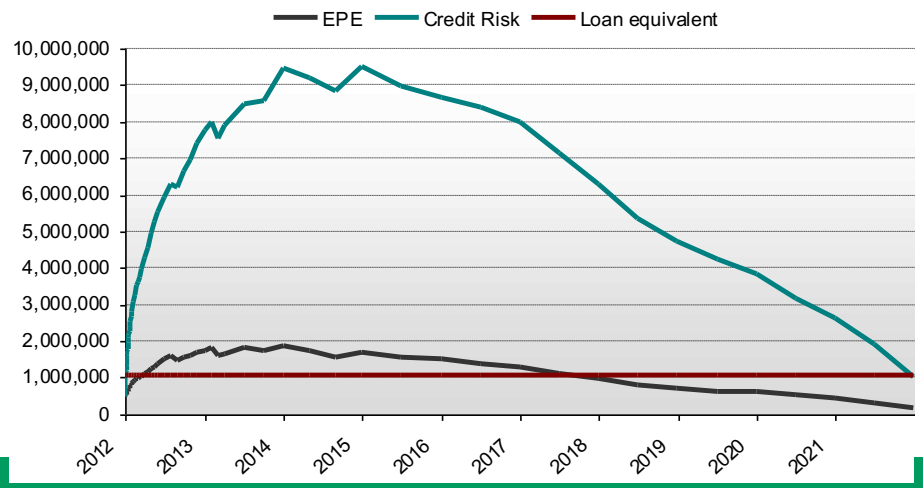
Non standard agreements

- **No CSA**
- **Unilateral CSA** (can be unilateral in our favour; often supranational institutions require an unilateral CSA in their favour)
- **CSA with very high threshold** – 50-100M threshold will impact pricing
- **CSA with rating triggers**
- **Sub-optimal CSAs** – negative spreads on cash collateral, securities received that can't be re-hypothecated

CVA – Impact of Collateral on IRS

- **The Credit Support Annex is a key tool for credit risk mitigation**
  - EPE computation includes netting and collateral agreements
  - EPE is calculated up to the threshold
  - Above threshold, mark to market drift is calculated on margin call period + collateral lag period (10days)
- **CSA features are built into EPE simulations and consequently have an impact on CVA**

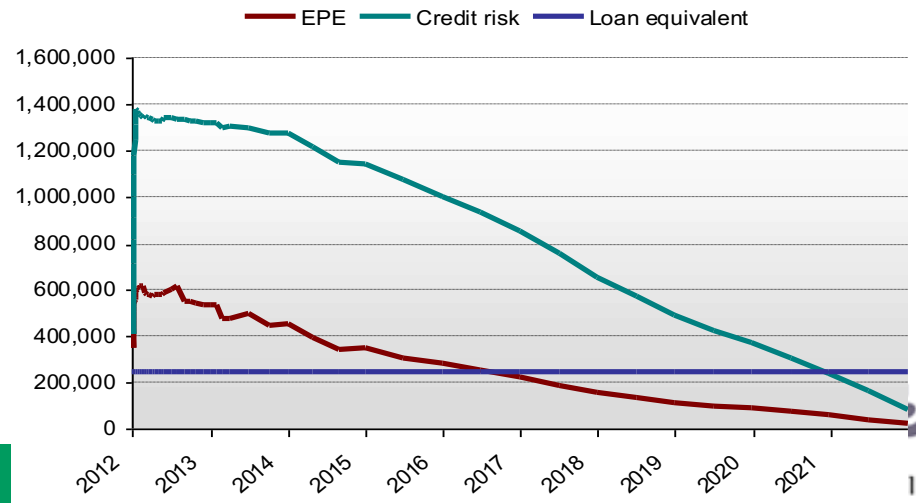
- **No collateral agreement in place**
  - Risk exposure is maximal for CA-CIB



- **Trade description**

Start date:	10 Apr 2012
Maturity:	10Y
Notional:	EUR 100M
CA-CIB receives:	2.30% (SA, act/360)
CA-CIB pays:	6M Euribor (SA, 30/360

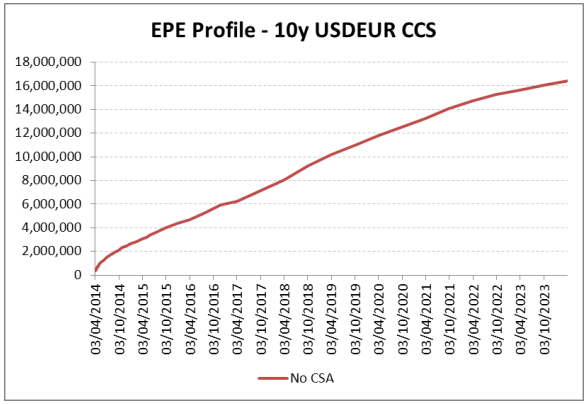
- **CSA agreement in place**
  - Unilateral vs Bilateral / Threshold / Minimum Transfer Amount (MTA)
  - Sharp decrease of risk profile due to CSA risk mitigation



CVA – Portfolio Diversification

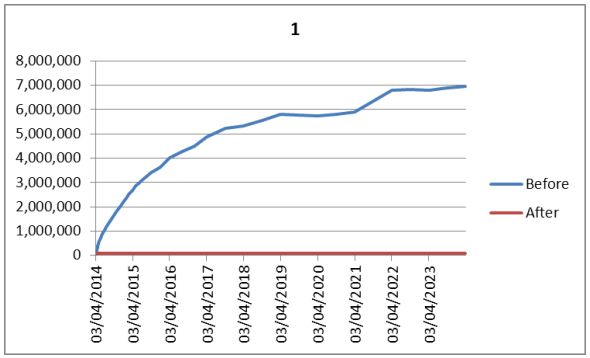
- **CVA is calculated at the netting set level - including diversification benefits within the existing portfolio.**

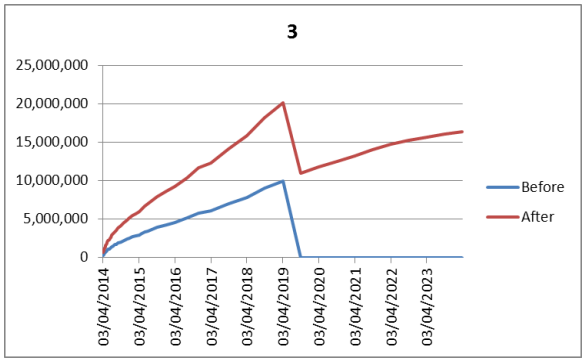
- **Example – Adding a 10y CCS (Pay USD fixed versus Receive EUR Fixed) to an existing portfolio of:**
  - 10y CCS (Receive USD fixed versus Pay EUR Fixed) – i.e. opposite direction
  - 5y CCS (Receive USD fixed versus Pay EUR Fixed) – i.e. opposite direction
  - 5y CCS (Pay USD fixed versus Receive EUR Fixed) – i.e. same direction
  - 10y EUR IRS (Pay EUR fixed, Receive floating)

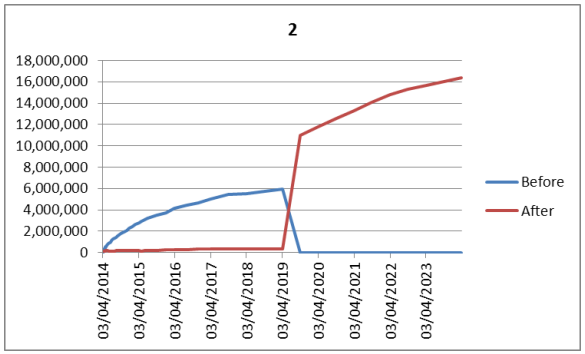


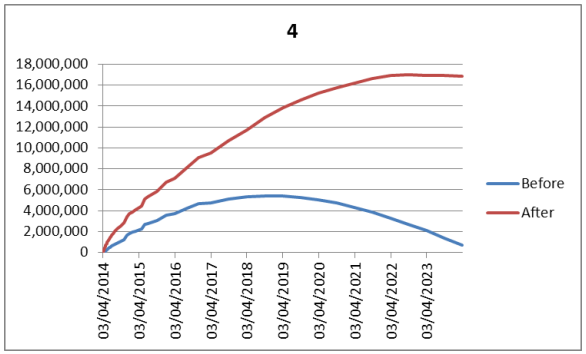
	Standalone
Before	0
After	3,391,762
Marginal	3,391,762

	1	2	3	4
Before	1,782,388	728,006	1,018,597	1,302,337
After	0	2,395,549	4,410,266	4,269,540
Marginal	-1,782,388	1,667,543	3,391,669	2,967,203









CRÉDIT AGRICOLE CORPORATE & INVESTMENT BANK

Summary of definitions

EPE : Expected Positive Exposure. Average of the MtM^+ obtained via Monte Carlo Simulation of a deal / portfolio: (all positive amounts) / (total number of scenarios) It can be viewed as the average risk of the portfolio. Fundamental measure to price and calculate XVA's.

CVA : Adjustment made to account for the credit risk embedded in a derivative transactions Probability of Default x Exposure (EPE) x Loss Given Default.

LVA / FVA : Liquidity / Funding Value Adjustment (LVA) is an adjustment made to the price of a derivative transaction to be consistent with the actual cost of funding it. LVA applies to counterparties with a non-standard CSA or no CSA in place.

Credit Support Annex (CSA): Legal document which regulates collateral for derivatives transactions. The main components are the threshold, the Minimum Transfer Amount (MTA), the eligible currency and collateral, and the collateral remuneration.

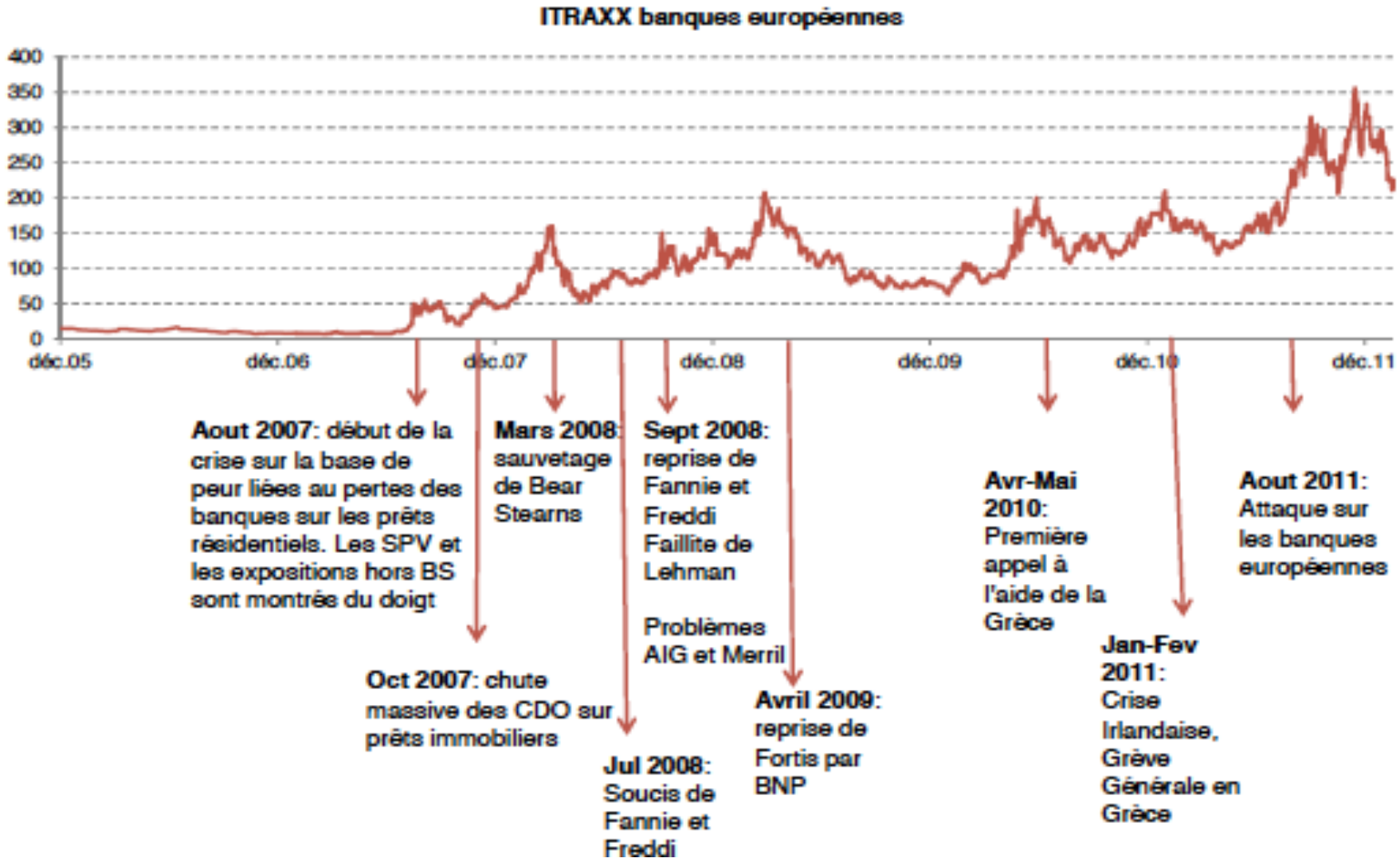
KVA : Capital Valuation adjustment. Adjustment made to the price of a derivative to reflect the cost of capital required by regulators which should be remunerated so that the deal reached the profitability requested by shareholders.

RWA : Risk Weighted Assets. Assets and off-balance-sheet exposures of the Bank, weighted according to risk, that is used to set the minimum amount of capital required by regulators.

Plan du cours

1. **Produits Structurés (Suite)**
   - Wrappers
2. **Notions sur le MARCHÉ DU CREDIT**
   - Le Risque de crédit
   - Credit Linked Note
   - Repackaged Note
3. **Le risque de crédit dans les Dérivés**
4. **Exemple de distorsions de marché en temps de crise**
   - La crise des dettes souveraines
   - La gestion des ressources rares

CHRONOLOGIE DE LA CRISE ET LE RISQUE BANCAIRE



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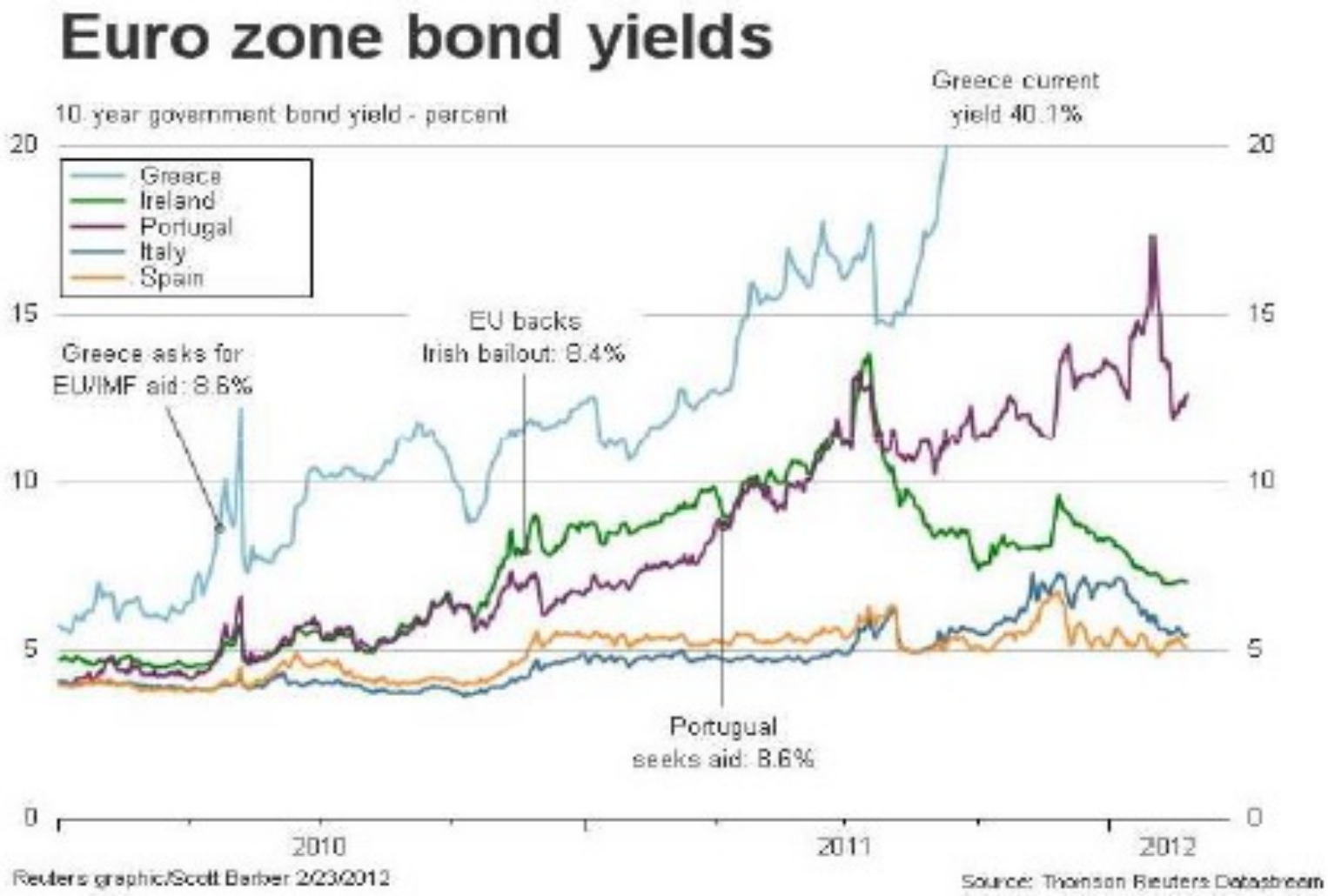
LA CRISE DES DETTES SOUVERAINES



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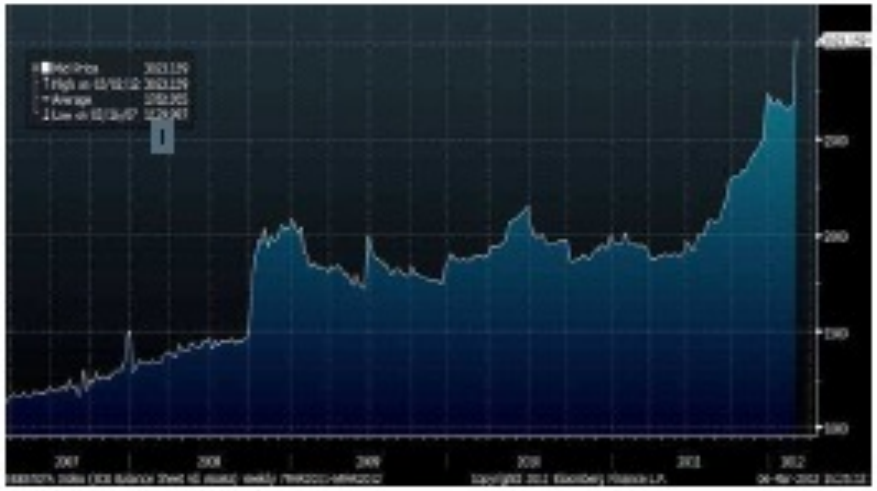
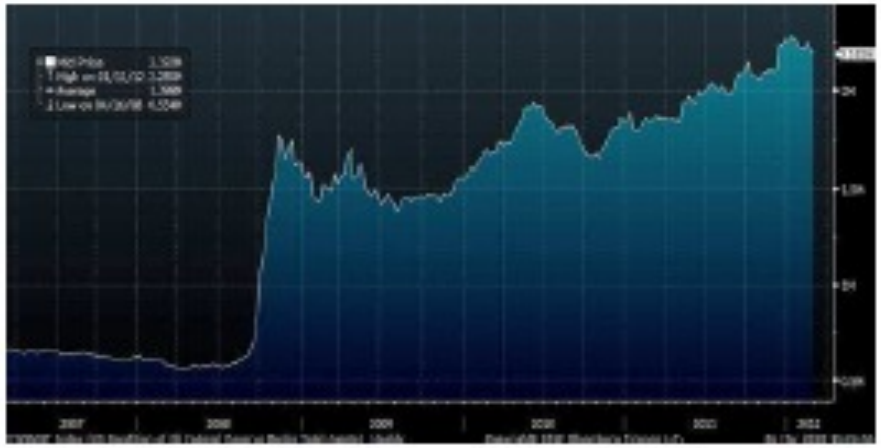
UNE CRISE DE LIQUIDITE

- **Les deux graphiques montrent le bilan de la banque centrale Européenne (en haut) et de la FED (en Bas), en EURO.**

- **La difficulté des banques à se financer, la crise de liquidité, à eu de nombreuses conséquences :**
  - Intervention des banques centrales en préteurs de dernier recours
  - La liquidité est devenue un actif en soi : un actif fundé traite à un prix différent du dérivé correspondant.
  - Repricing global de toutes les position fundées, y compris de la position de funding dans les opérations dérivés
  - Par exemple dans les dérivés sur part de fond, sur action, ou dans les dérivés non collatéralisés





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CONSEQUENCE : LIBOR-OIS SPREAD

- La base LIBOR-OIS est un bon indicateur des tensions
- Lors d'une crise de crédit, les banques préfèrent prêter à 1 jour qu'à 1m, 3m ou 6m.
- De ce fait les libors, qui représentent les couts d'emprunts bancaires à moyen terme deviennent plus élevés que les taux au jour le jour.

To follow it on bloomberg, type CMW, then choose your region

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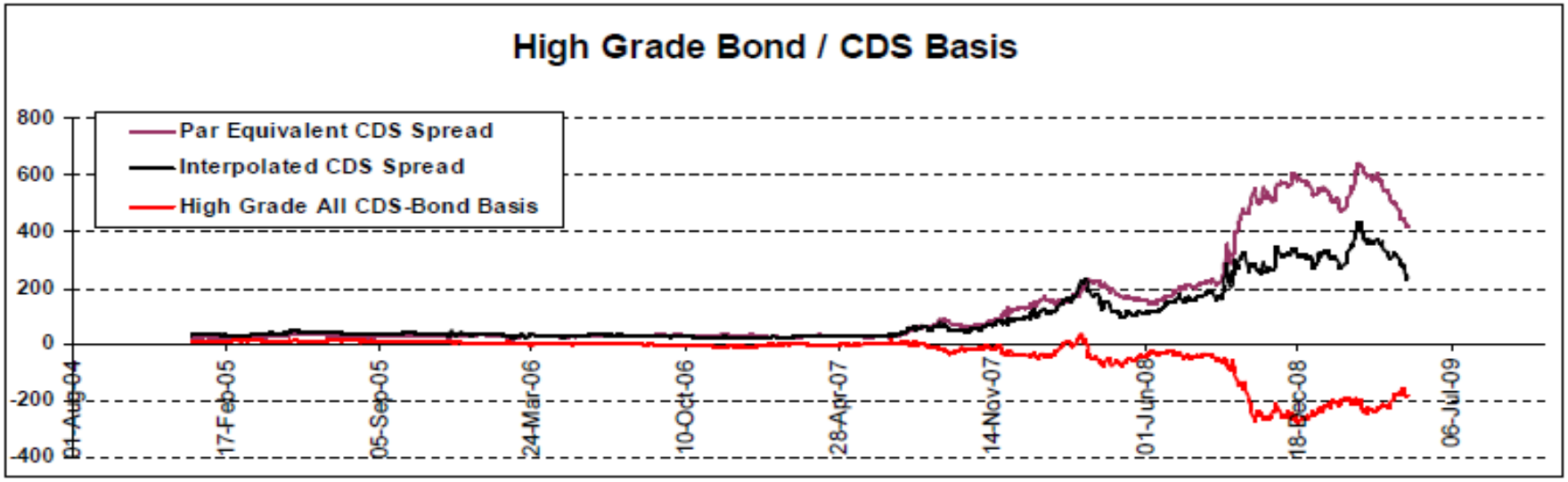
CONSEQUENCE : L'APPARITION DES BASES NEGATIVES

- **De la même manière CDS et Bonds ne sont plus liés directement en cas de crise**
  - Manque de liquidité
  - Faible liquidité des obligations
  - Baisse du nombre de preneurs de risques



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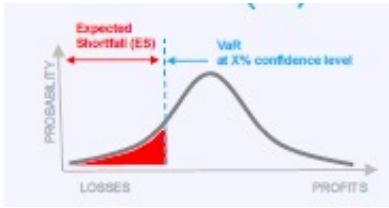
Tougher reglementation as a response to the Crisis

- **The present (Basel 3)**
  - Higher capital ratio … but less eligible capital
  - More RWA

- **Fondamental Review of the Trading Book (2018)**
  - Shift from VaR/sVaR to expected shortfall to captures the 'tail risk' i.e. extreme losses beyond the VaR confidence level
  - Liquidity Horizons differentiated by risk factors (ranging from 10 days to 120 days)
  - Internal model approval at trading desk level (Backtesting and P&L attribution are conducted at the trading desk level)
  - New standard approach used as a floor to internal models



Daily VaR x $\sqrt{10}$ x scaling factor

- **Higher Liquidity requirements**
  - LCR : to face a 30 days liquidity stress scenario
  - NSFR : to face a 1Y closure of funding wholesale market

- **Constrains on the use of the Balance Sheet**
  - Leverage ration

$$\text{Leverage ratio} = \frac{\text{Additional T1}}{\text{On-B\&S} \quad \text{Off-B\&S}} \geq 3\%$$

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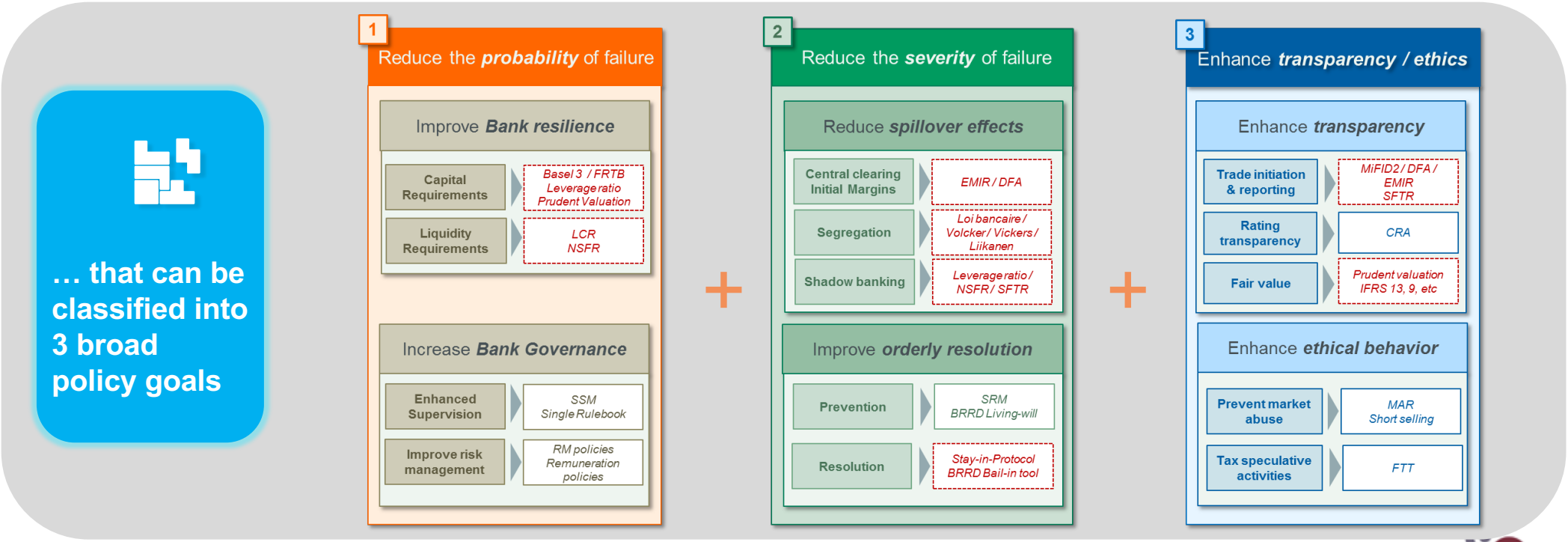
A huge evolution of the banking regulatory landscape



SSM : Single Supervisory Mechanism / SRM : Single Resolution Mechanism / BRRD: Banking Recovery and Resolution Directive / SFTR: Security Financing Transactions Reg. / CRA: Credit Rating Agencies / MAR: Market Abuse Reg.

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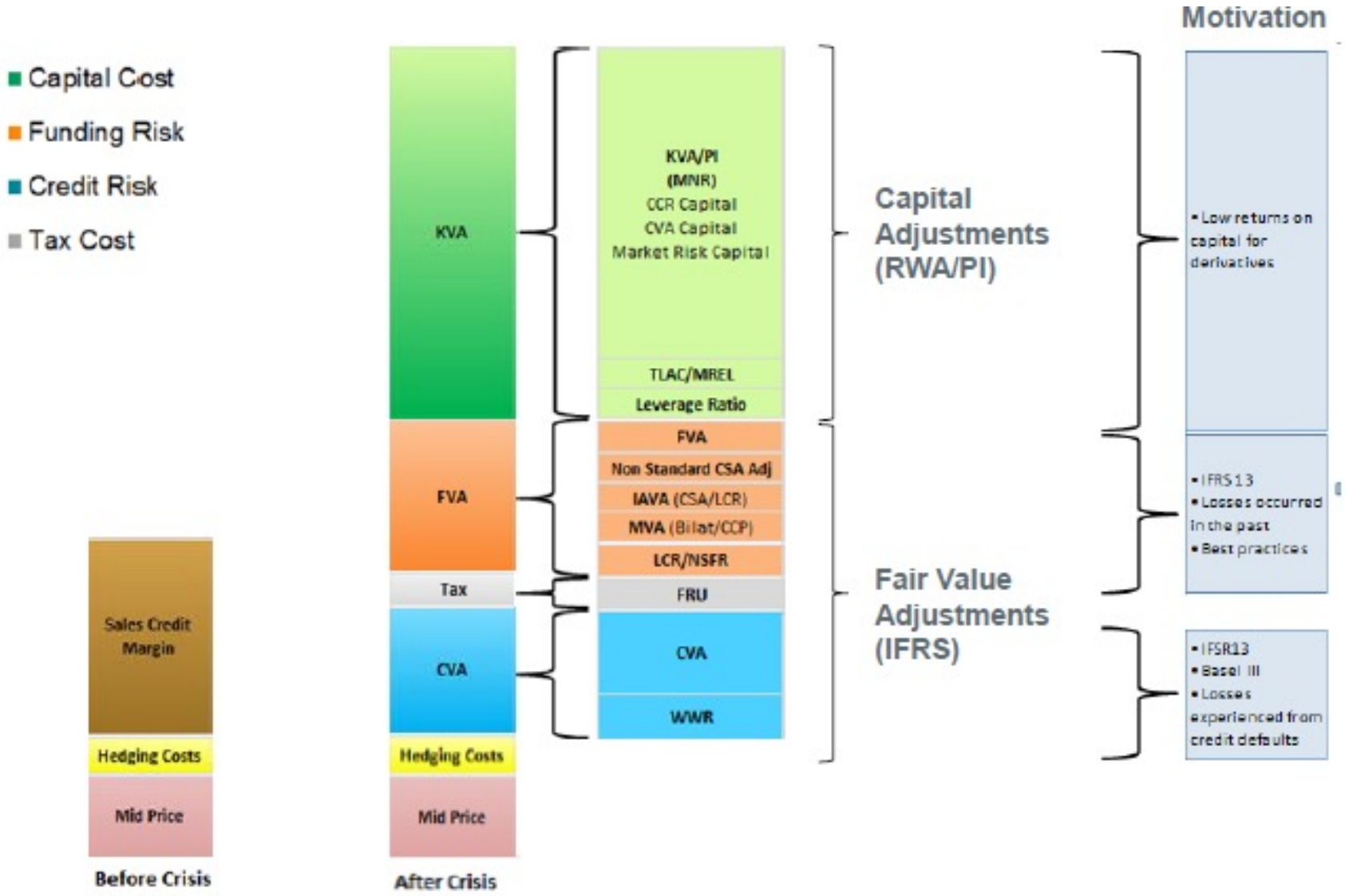
XVA Pricing Breakdown

From a simple definition to a much more complex one



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XVA Pricing Breakdown

From a simple definition to a much more complex one

- **Before the Crisis**, the Sales Credit Margin encompassed partially credit and funding risk.
- As it was entirely recognised as NBI, Sales were incentivised to deal riskier transactions because their Credit and Funding risk was generating more NBI.

- **After the Crisis**, the regulator asked Banks to provision Credit and Funding risk under the denomination “xVAs”.
- Derivatives prices have risen and breakdown became more complex.
- Due to the non-simultaneous implementation of these new metrics, biddings are more competitive.
- Some of these metrics are portfolio dependent allowing for netting effects and optimisation.

XVA role in Pricing

- Understanding XVA does not need a quantitative background, but understanding the concepts is very important to being able to interpret the costs and to optimise their values.

$$PV_{risky} = PV_{risk\ free}$$

$-CVA + DVA$	← Counterparty Risk
$\pm FVA$	← Funding Costs
$\pm ColVA$	← Non-standard Collateral Costs
$\pm RVA$	← Replacement Costs
$-KVA$	← Cost of Capital
$-AVA$	← Prudential Valuation
$-MVA$	← Initial Margin Adjustment

- A derivatives trade not taking into account its XVA is essentially **mispriced**!

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